4. Scenario: You are working on a project that involves analyzing the sales performance of a company over the past four quarters. The quarterly sales data is stored in a NumPy array named sales_data, where each element represents the sales amount for a specific quarter. Your task is to calculate the total sales for the year and determine the percentage increase in sales from the first quarter to the fourth quarter. Question: Using NumPy arrays and arithmetic operations calculate the total sales for the year and determine the percentage increase in sales from the first quarter to the fourth quarter?

Code;

```
import numpy as np
# Quarterly sales data
sales_data = np.array([10000, 12000, 15000, 20000])
# Calculate total sales
total_sales = np.sum(sales_data)
# Calculate percentage increase from Q1 to Q4
percentage_increase = (
    (sales_data[3] - sales_data[0]) / sales_data[0]
) * 100

# Display results

print("Quarterly Sales:", sales_data)

print("Total Yearly Sales:", total_sales)

print("Percentage Increase:", percentage_increase, "%")
```

INPUT;

exxxxxxxxxxpppp4 fds.py - C:/Users/Jashw/exxxxxxxxxxpppp4 fds.py (3.13.14)

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```
import numpy as np

# Quarterly sales data
sales_data = np.array([10000, 12000, 15000, 20000])

# Calculate total sales
total_sales = np.sum(sales_data)

# Calculate percentage increase from Q1 to Q4
percentage_increase = (
    (sales_data[3] - sales_data[0]) / sales_data[0]
) * 100

# Display results
print("Quarterly Sales:", sales_data)
print("Total Yearly Sales:", total_sales)
print("Percentage Increase:", percentage_increase, "%")
```

OUTPUT;

```
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>
================ RESTART: C:/Users/Jashw/exxxxxxxxxxpppp4 fds.py ===============
Quarterly Sales: [10000 12000 15000 20000]
Total Yearly Sales: 57000
Percentage Increase: 100.0 %
>
```